Ordinary Differential Equations

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1 Definitions

Definition 1 (Ordinary differential equation)**.** *Let $U \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and let $f : U \to \mathbb{R}^n$ be continuous. A* first-order ordinary differential equation *(ODE) on U is the equation*

$$y'(t) = f(t, y(t)),$$

where $y : I \to \mathbb{R}^n$ is an unknown differentiable function defined on an interval $I \subseteq \mathbb{R}$ with $\{(t, y(t)) : t \in I\} \subseteq U$. The function f is called the vector field *of the equation.*

Definition 2 (Initial value problem)**.** *Given $(t_0, y_0) \in U$, the* initial value problem *(IVP) is the pair*

$$y'(t) = f(t, y(t)), \qquad y(t_0) = y_0.$$

A solution *is a differentiable map $y : I \to \mathbb{R}^n$ with $t_0 \in I$ satisfying both equations.*

2 The canonical example

Theorem 1 (Solution of $y' = -y$, $y(0) = 1$)**.** *Let $f(t, y) = -y$. The unique solution $y : \mathbb{R} \to \mathbb{R}$ of the IVP*

$$y'(t) = -y(t), \qquad y(0) = 1$$

is

$$y(t) = e^{-t}.$$

Proof. **Existence (direct substitution).** Define $y(t) := e^{-t}$. Then y is differentiable on $\mathbb{R}$ with

$$y'(t) = \frac{d}{dt}e^{-t} = -e^{-t} = -y(t),$$

so y satisfies the ODE. Moreover $y(0) = e^0 = 1$, so the initial condition holds. Hence $y(t) = e^{-t}$ is a solution.

Uniqueness. Suppose $\tilde{y} : \mathbb{R} \to \mathbb{R}$ is any solution. Define $g(t) := e^t \tilde{y}(t)$. Then g is differentiable with

$$g'(t) = e^t \tilde{y}(t) + e^t \tilde{y}'(t) = e^t \tilde{y}(t) + e^t(-\tilde{y}(t)) = 0.$$

So g is constant on $\mathbb{R}$. Evaluating at $t = 0$: $g(0) = e^0 \tilde{y}(0) = 1$. Thus $e^t \tilde{y}(t) \equiv 1$, i.e. $\tilde{y}(t) = e^{-t}$ for all $t \in \mathbb{R}$.

The two parts together prove existence and uniqueness, so $y(t) = e^{-t}$ is the unique solution. □

Lemma 2 (Picard–Lindelöf, scalar version)**.** *If $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is continuous and Lipschitz in its second argument, then the IVP $y'(t) = f(t, y(t))$, $y(t_0) = y_0$ has a unique solution on a neighborhood of t_0.*

The vector field $f(t, y) = -y$ is globally Lipschitz with constant 1, so Picard–Lindelöf applies and gives an independent proof of uniqueness in Theorem 1.

3 Remark

The solution operator $y_0 \mapsto e^{-t}y_0$ is the prototype of a *linear semigroup*: it is the one-parameter family $\{\Phi_t\}_{t\geq 0}$ with $\Phi_t = e^{-t}$, $\Phi_0 = \mathrm{id}$, $\Phi_{s+t} = \Phi_s\Phi_t$. Generalizing this construction to operator-valued A yields e^{tA} and the entire theory of constant-coefficient linear ODEs, which underlies the analysis of stability of equilibria in dynamical systems and learned vector fields in flow-matching generative models.